Lab 7 - Fuzz Testing

SRN: PES2UG24CS036
Date: 8 November 2026

Repository: https://github.com/SETAPESU26/L6_L7_33_spam_scoring

Aim

To use Hypothesis to find input validation bugs in the spam scoring program and fix them.

1. Running the original program

`python spam_scoring.py`

The sample inputs worked, but leaving the email blank caused an error.

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 01_before_normal | Exit code: 0
Command: python spam_scoring.py
Working directory: /home/adityashrivastav/SE_Lab/Lab-7/original
stdin (escaped): "offers@Deals.com\nVisit https://example.com/deal now\nFREE prize NOW\n"
Complete stdout and stderr from the recorded subprocess, displayed verbatim below.

```
==== Running sender_domain ====
Enter a sender email (e.g. offers@Deals.com): Sender Domain: deals.com

==== Running first_link ====
Enter a message containing a link: First Link: https://example.com/deal

==== Running shout_ratio ====
Enter a message (e.g. FREE prize NOW): Uppercase Ratio: 0.5833333333333334
```

Output with a valid email, a message containing a link, and normal text.

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 02_before_invalid | Exit code: 1
Command: python spam_scoring.py
Working directory: /home/adityashrivastav/SE_Lab/Lab-7/original
stdin (escaped): "\n\n\n"
Complete stdout and stderr from the recorded subprocess, displayed verbatim below.

```
==== Running sender_domain ====
Enter a sender email (e.g. offers@Deals.com): Traceback (most recent call last):
  File "/home/adityashrivastav/SE_Lab/Lab-7/original/spam_scoring.py", line 33, in <module>
    main()
    ~~~~^^
  File "/home/adityashrivastav/SE_Lab/Lab-7/original/spam_scoring.py", line 19, in main
    result = sender_domain(user_input)
  File "/home/adityashrivastav/SE_Lab/Lab-7/original/spam_scoring.py", line 2, in sender_domain
    return email.split('@')[1].lower()
           ~~~~~~~~~~~~~~~~^^^
IndexError: list index out of range
```

A blank email caused IndexError because the code directly accessed the second part after splitting at @.

2. Fuzz tests before fixing the code

The tests use @given to generate strings, None, numbers and other input types. The first run failed for all three functions with None as the input.

```
python -m pytest test_discovery.py -q -k "not text_no_crash"
```

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 03_before_fuzz | Exit code: 1
Command: python -m pytest test_discovery.py -q -k not text_no_crash --tb=short --color=no
Working directory: /home/adityashrivastav/SE_Lab/Lab-7/original
Complete stdout and stderr from the recorded subprocess, displayed verbatim below.

```
FFF                                                                      [100%]
================================== FAILURES ===================================
_____________________________ test_sender_domain_no_crash _____________________________
test_discovery.py:14: in test_sender_domain_no_crash
    @given(INPUTS)

test_discovery.py:16: in test_sender_domain_no_crash
    result = sender_domain(value)
             ^^^^^^^^^^^^^^^^^^^^
spam_scoring.py:2: in sender_domain
    return email.split('@')[1].lower()
           ^^^^^^^^^^^
E   AttributeError: 'NoneType' object has no attribute 'split'
E   Failing test case: test_sender_domain_no_crash(
E       value=None,
E   )
_____________________________ test_first_link_no_crash _____________________________
test_discovery.py:22: in test_first_link_no_crash
    @given(INPUTS)

test_discovery.py:24: in test_first_link_no_crash
    result = first_link(value)
             ^^^^^^^^^^^^^^^^^
spam_scoring.py:5: in first_link
    start = text.index('http')
            ^^^^^^^^^^
E   AttributeError: 'NoneType' object has no attribute 'index'
E   Failing test case: test_first_link_no_crash(
E       value=None,
E   )
_____________________________ test_shout_ratio_no_crash _____________________________
test_discovery.py:30: in test_shout_ratio_no_crash
    @given(INPUTS)

test_discovery.py:32: in test_shout_ratio_no_crash
    result = shout_ratio(value)
             ^^^^^^^^^^^^^^^^^^
spam_scoring.py:12: in shout_ratio
    letters = [c for c in text if c.isalpha()]
                          ^^^^
E   TypeError: 'NoneType' object is not iterable
E   Failing test case: test_shout_ratio_no_crash(
E       value=None,
E   )
=========================== short test summary info ============================
FAILED test_discovery.py::test_sender_domain_no_crash - AttributeError: 'None...
FAILED test_discovery.py::test_first_link_no_crash - AttributeError: 'NoneTyp...
FAILED test_discovery.py::test_shout_ratio_no_crash - TypeError: 'NoneType' o...
3 failed, 3 deselected in 0.16s
```

sender_domain and first_link raised AttributeError. shout_ratio raised TypeError. The other three tests were not selected in this run.

3. Testing empty strings

The text-only tests also failed. Hypothesis reduced the failing input to an empty string in all three cases.

```
python -m pytest test_discovery.py -q -k text_no_crash
```

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 03b_before_text_fuzz | Exit code: 1
Command: python -m pytest test_discovery.py -q -k text_no_crash --tb=short --color=no
Working directory: /home/adityashrivastav/SE_Lab/Lab-7/original
Complete stdout and stderr from the recorded subprocess, displayed verbatim below.

```
FFF                                                                      [100%]
================================== FAILURES ===================================
_____________________ test_sender_domain_text_no_crash ______________________
test_discovery.py:38: in test_sender_domain_text_no_crash
    @given(st.text())
                ^^^
test_discovery.py:40: in test_sender_domain_text_no_crash
    assert isinstance(sender_domain(value), str)
                      ^^^^^^^^^^^^^^^^^^^^
spam_scoring.py:2: in sender_domain
    return email.split('@')[1].lower()
           ^^^^^^^^^^^^^^^^^^
E   IndexError: list index out of range
E   Failing test case: test_sender_domain_text_no_crash(
E       value='',
E   )
_____________________ test_first_link_text_no_crash _____________________
test_discovery.py:44: in test_first_link_text_no_crash
    @given(st.text())
                ^^^
test_discovery.py:46: in test_first_link_text_no_crash
    assert isinstance(first_link(value), str)
                      ^^^^^^^^^^^^^^^^^^
spam_scoring.py:5: in first_link
    start = text.index('http')
            ^^^^^^^^^^^^^^^^^^
E   ValueError: substring not found
E   Failing test case: test_first_link_text_no_crash(
E       value='',  # or any other generated value
E   )
_____________________ test_shout_ratio_text_no_crash _____________________
test_discovery.py:50: in test_shout_ratio_text_no_crash
    @given(st.text())
                ^^^
test_discovery.py:52: in test_shout_ratio_text_no_crash
    assert 0.0 <= shout_ratio(value) <= 1.0
                  ^^^^^^^^^^^^^^^^^^
spam_scoring.py:14: in shout_ratio
    return len(upper) / len(letters)
           ^^^^^^^^^^^^^^^^^^^^^^^^^
E   ZeroDivisionError: division by zero
E   Failing test case: test_shout_ratio_text_no_crash(
E       value='',
E   )
=========================== short test summary info ============================
FAILED test_discovery.py::test_sender_domain_text_no_crash - IndexError: list...
FAILED test_discovery.py::test_first_link_text_no_crash - ValueError: substri...
FAILED test_discovery.py::test_shout_ratio_text_no_crash - ZeroDivisionError:...
3 failed, 3 deselected in 0.17s
```

The errors were IndexError in sender_domain, ValueError in first_link, and ZeroDivisionError in shout_ratio.

4. Fixing and running the program again

```
python spam_scoring.py
```

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 04_after_normal | Exit code: 0
Command: python spam_scoring.py
Working directory: /home/adityashrivastav/SE_Lab/Lab-7
stdin (escaped): "offers@Deals.com\nVisit https://example.com/deal now\nFREE prize NOW\n"
Complete stdout and stderr from the recorded subprocess, displayed verbatim below.

```
==== Running sender_domain ====
Enter a sender email (e.g. offers@Deals.com): Sender Domain: deals.com

==== Running first_link ====
Enter a message containing a link: First Link: https://example.com/deal

==== Running shout_ratio ====
Enter a message (e.g. FREE prize NOW): Uppercase Ratio: 0.5833333333333334
```

The valid sample inputs still gave the same output after the changes.

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 05_after_invalid | Exit code: 0
Command: python spam_scoring.py
Working directory: /home/adityashrivastav/SE_Lab/Lab-7
stdin (escaped): "\n\n\n"
Complete stdout and stderr from the recorded subprocess, displayed verbatim below.

```
==== Running sender_domain ====
Enter a sender email (e.g. offers@Deals.com): Sender Domain:

==== Running first_link ====
Enter a message containing a link: First Link:

==== Running shout_ratio ====
Enter a message (e.g. FREE prize NOW): Uppercase Ratio: 0.0
```

With three blank inputs, the program now completes. It returns empty strings for the domain and link, and 0.0 for the ratio.

Changes made

sender_domain: Check the input type and address format before splitting it. Return an empty string if the address is invalid.

first_link: Check for an HTTP or HTTPS link instead of assuming one exists. Return an empty string if there is no valid link.

shout_ratio: Check whether any letters are present before dividing. Return 0.0 for empty input or text with no letters.

5. Re-running the tests

The same six tests that failed earlier passed after the fixes.

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 06_after_discovery | Exit code: 0
Command: python -m pytest test_spam_scoring.py -k no_crash -q --tb=short --color=no
Working directory: /home/adityashrivastav/SE_Lab/Lab-7
Complete stdout and stderr from the recorded subprocess, displayed verbatim below.

```
......                                                                   [100%]
6 passed, 66 deselected in 0.90s
```

All six discovery tests passed. The additional tests were not selected in this run.

`python -m pytest test_spam_scoring.py -v`

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 07_after_full_suite | Exit code: 0
Command: python -m pytest test_spam_scoring.py -v --tb=short --color=no --hypothesis-show-statistics
Working directory: /home/adityashrivastav/SE_Lab/Lab-7
Output excerpt: final 14 lines. Complete output is preserved in evidence/logs/07_after_full_suite.txt

```
  - Stopped because settings.max_examples=500


test_spam_scoring.py::test_nonletters_do_not_change_shout_ratio:

  - during generate phase (0.14 seconds):
    - Typical runtimes: < 1ms, of which < 1ms in data generation
    - 500 passing, 0 failing, and 0 invalid test cases

  - Stopped because settings.max_examples=500


============================== 72 passed in 1.89s ==============================
```

Final test result: 72 passed. The screenshot shows the end of the full test output.

Result

All 72 tests passed. There are 13 Hypothesis tests with 500 generated examples each, and 59 fixed test cases. The extra cases check malformed addresses, missing links, Unicode text, whitespace and long inputs.

The main issue was that the original functions assumed the input would always be valid. Fuzzing found failures with None and empty strings that the normal sample inputs did not show.

The fixed code is in spam_scoring.py and the tests are in test_spam_scoring.py. The screenshots display saved command output in a browser. The full logs are in evidence/logs/.

Submission: https://github.com/Audatic07/SE_Lab/tree/main/Lab-7